CardioCare Quest

Tyler Jeffrey, Jesse Ceja, Anna Cheatham, Rique Yazzie

Mentor: Morgan Nicholson

Elevator Pitch / Introduction

Hypertension is globally estimated to affect **1.4 billion adults**, with only **25% of that population having it under control**. This problem disproportionately affects the indigenous communities, facing prevalence rates that are **10-20% percent higher** than non-indigenous groups as a result of systemic healthcare inequalities and cultural mismatches in medical intervention.

Our client, **Dr. Jared Duval**, aims to address this issue with CardioCare Quest, an NIH funded platform that replaces the isolating and “clinical” feel of current health apps to encourage a culturally grounded form of play. Rather than a simple tracker, CardioCare Quest allows users to manage heart health by participating in **culturally-relevant mobile health games**.

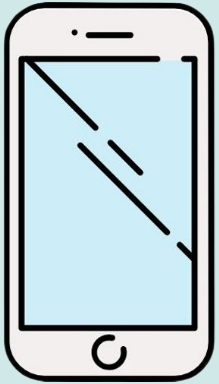
Our role in this project is to build the **Twine-integrated app shell**, allowing users to “write” their own cultural narratives directly into the game for others to play. Additionally, we will be **designing a number of games** based on workshops conducted alongside community members. This app lays the foundation for future interventions that are culturally resonant and community-led alongside reducing health disparities for groups that have historically been underserved by both healthcare and technology.

CardioCare Quest Docs

Introduction

Hypertension

- Affects 1.4 billion adults
- 25% under control
- 10%-20% higher rates with indigenous people



Mobile Health Game

- Replaces isolated and "clinical" feeling of health apps
- Culturally relevant

Twine Integration

- Users can "write" their own cultural narratives
- Designing a number of games

Client Introduction

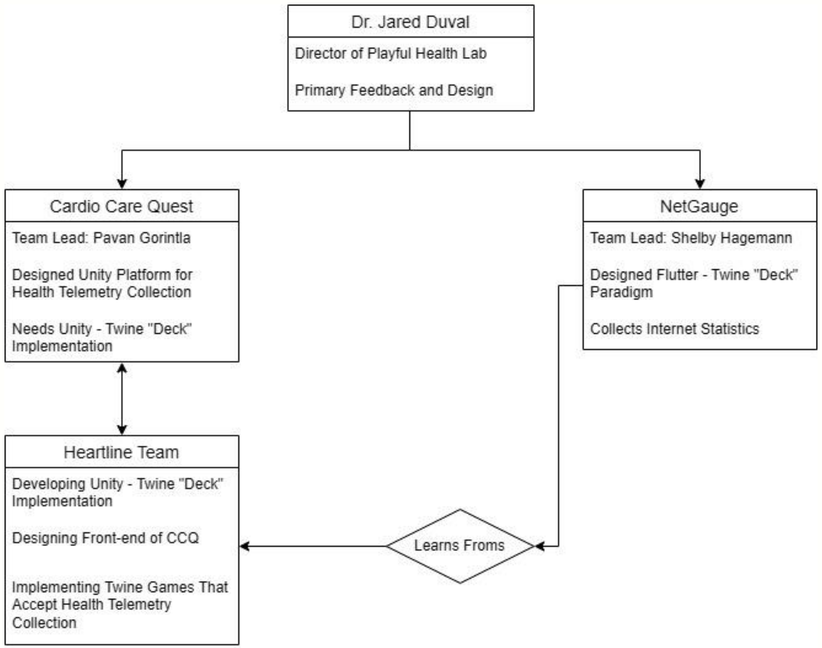
Dr. Jared Duval

- Director of Playful Health Technology Lab.
- Assistant Professor in School of Informatics, Computing, and Cyber Systems.
- Bridges between Human-Computer interactions with people’s health, focusing on Games and Play.

By connecting people’s enjoyment with tasks that improve their daily health maintenance, Dr. Duval makes people enjoy the process of building healthy habits.

The Playful Health Technology Lab

- **Dr. Duval Lead**
  - Everything comes down from Duval
  - Head of Feedback, assigns other team members peer-based feedback
- **Phase Based Development**
  - Testing, Design, Development
  - Testing underscores it all
- **“Next 5” Style Planning**
- **Proof Of Success Points**



The Core Issues

With current barriers in existing landscapes of health technology, CardioCare Quest comes to address these **four** limitations of current health apps:

1. Isolating Design

"Disease Centered", focusing on data input and self monitoring rather than social connection.

2. Clinical Alienation

Western healthcare systems create skepticism causing mistrust in patients, causing isolation.

3. Static Content

Current apps aren't designed to allow for communal input & interaction, focusing on data management

4. Cultural Disparities

Structural and cultural barriers like travel distances between healthcare providers and isolation of rural communities exacerbate healthcare access

Solution

CardioCare Quest aims to **gamify** the patient journey:

1 **Narrative-Based Motivation**
Uses storytelling to increase engagement

2 **Interactive "Quests"**
Story missions give an opportunity to learn, diet, and exercise.

3 **Community Authorship**
Users will be able to create and publish their own narratives.

Core Platform Features

Twine Integration

The bridge between Twine and the Unity platform for clinical data collection and game creation.

Cohesive App Shell

A unified system for providing seamless navigation between various game styles.

Plan For Development

Phase Based Development

Phase 1: Design and Testing	Phase 2: Development and Testing
• Iterative “Figma” design • Scheduled Meetings with Gorintla for identifying met and unmet design requirements.	• Cross referencing Hagemann's design to develop our Twine-Unity bridge. • “Next 5” planning to foster rapid weekly prototyping • Gorintla’s feedback to ensure lacking core functionality is identified early

Issue: As of now, Cardio Care Quest only has permission to directly design for the Navajo Nation and cannot specify for any other Tribal Communities.

Technical Investigation

Core Integration & Platform Mechanics

1 Twine-Unity Bridge

Researching how the NetGauge Twine "deck" paradigm was implemented and integrate.

2 Unity Platform Accommodations

Cross-referencing technical documentation to assist our integration with client's existing Health Telemetry collection platform

User Experience & System Connectivity

3 UI/UX Shell Design

Refining navigation by cross-referencing existing design layouts and research workshops.

4 Mobile OS Based Connections

Investigating platform specific APIs and hardware connections to ensure proper data collection across devices.

In Conclusion,

With the guidance of the PHT lab, we will be:

- Gamifying the patient journey for indigenous people, specifically the Navajo community.
- Implementing solutions to the limitations of current medical health apps.
- Integrating Twine with Unity to make simple yet effective games.

Ahéhee'

T'aa anołtso